

Skill Session 9

Task 1

To Change Directories, Validate Paths, Update Working Directory, Handle Errors, Maintain Previous Directory, Test Navigation

```
j_rakshitha@RAKSHITHA:~$ cd ~/ossp_project
j_rakshitha@RAKSHITHA:~/ossp_project$ pwd
/home/j_rakshitha/ossp_project
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/session9_task1_cd.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/session9_task1_cd.c -o build/session9_task1_cd
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/session9_task1_cd
=== Session 9 Task 1: Directory Navigation ===
Current directory: /home/j_rakshitha/ossp_project
cd-shell> cd-shell> pwd
Invalid command: cd-shell>
Current directory: /home/j_rakshitha/ossp_project
cd-shell> pwd
Current directory: /home/j_rakshitha/ossp_project
Current directory: /home/j_rakshitha/ossp_project
cd-shell> cd /tmp
Changed directory to: /tmp
Current directory: /tmp
cd-shell> pwd
Current directory: /tmp
Current directory: /tmp
cd-shell> cd ..
Changed directory to: ..
Current directory: /
cd-shell> pwd
Current directory: /
Current directory: /
cd-shell> cd
Changed directory to: /home/j_rakshitha
Current directory: /home/j_rakshitha
cd-shell> cd /tmp
Changed directory to: /tmp
Current directory: /tmp
cd-shell> cd /home
Changed directory to: /home
Current directory: /home
cd-shell> cd -
Changed directory to: /tmp
Current directory: /tmp
```

```
Changed directory to: /tmp
Current directory: /tmp
cd-shell> cd -
Changed directory to: /home
Current directory: /home
cd-shell> cd-shell> cd /something_that_does_not_exist
Invalid command: cd-shell>
Current directory: /home
cd-shell> cd-shell> cd /etc/passwd
Invalid command: cd-shell>
Current directory: /home
cd-shell> cd /something_that_does_not_exist
cd: No such file or directory
Current directory: /home
cd-shell> cd /etc/passwd
cd: /etc/passwd is not a directory
Current directory: /home
cd-shell> exit
Exiting...
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task1_cd.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>
#include <limits.h>
#include <sys/stat.h>

#define PATH_SIZE PATH_MAX

void print_current_directory(void)
{
    char cwd[PATH_SIZE];

    if (getcwd(cwd, sizeof(cwd)) == NULL)
    {
        perror("getcwd");
        return;
    }

    printf("Current directory: %s\n", cwd);
}

int validate_directory(const char *path)
{
    struct stat info;

    if (stat(path, &info) != 0)
    {
        perror("cd");
        return 0;
    }

    if (!S_ISDIR(info.st_mode))
    {
        fprintf(stderr, "cd: %s is not a directory\n", path);
        return 0;
    }

    return 1;
}

int change_directory(const char *path, char *previous_dir,
                    size_t previous_size)
{
    char current_dir[PATH_SIZE];

    ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location
    ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~/o × + v
GNU nano 8.7.1 src/session9_task1_cd.c *
char current_dir[PATH_SIZE];

if (getcwd(current_dir, sizeof(current_dir)) == NULL)
{
    perror("getcwd");
    return 0;
}

/* Save current directory before changing */
strncpy(previous_dir, current_dir, previous_size - 1);
previous_dir[previous_size - 1] = '\0';

if (!validate_directory(path))
{
    return 0;
}

if (chdir(path) != 0)
{
    perror("cd");
    return 0;
}

return 1;
}

int main(void)
{
    char input[PATH_SIZE];
    char previous_dir[PATH_SIZE] = "";
    char *command;
    char *argument;

    printf("=== Session 9 Task 1: Directory Navigation ===\n");

    while (1)
    {
        print_current_directory();

        printf("cd-shell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;
    }
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task1_cd.c *
    break;

    input[strcspn(input, "\n")] = '\0';

    command = strtok(input, " \t");
    argument = strtok(NULL, " \t");

    if (command == NULL)
        continue;

    if (strcmp(command, "exit") == 0)
    {
        printf("Exiting...\n");
        break;
    }

    if (strcmp(command, "pwd") == 0)
    {
        print_current_directory();
        continue;
    }

    if (strcmp(command, "cd") == 0)
    {
        char target[PATH_SIZE];

        if (argument == NULL)
        {
            const char *home = getenv("HOME");

            if (home == NULL)
            {
                fprintf(stderr, "cd: HOME not set\n");
                continue;
            }

            strncpy(target, home, sizeof(target) - 1);
            target[sizeof(target) - 1] = '\0';
        }
        else if (strcmp(argument, "-") == 0)
        {
            if (previous_dir[0] == '\0')
            {
                fprintf(stderr, "cd: previous directory not set\n");
                continue;
            }
        }
    }

```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_ Replace	^U Paste	^J Justify	^_ Go To Line

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task1_cd.c *
{
    const char *home = getenv("HOME");

    if (home == NULL)
    {
        fprintf(stderr, "cd: HOME not set\n");
        continue;
    }

    strncpy(target, home, sizeof(target) - 1);
    target[sizeof(target) - 1] = '\0';
}
else if (strcmp(argument, "-") == 0)
{
    if (previous_dir[0] == '\0')
    {
        fprintf(stderr, "cd: previous directory not set\n");
        continue;
    }

    strncpy(target, previous_dir, sizeof(target) - 1);
    target[sizeof(target) - 1] = '\0';
}
else
{
    strncpy(target, argument, sizeof(target) - 1);
    target[sizeof(target) - 1] = '\0';
}

if (change_directory(target,
                    previous_dir,
                    sizeof(previous_dir)))
{
    printf("Changed directory to: %s\n", target);
}

continue;
}

printf("Invalid command: %s\n", command);
}

return 0;
}
|
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Task 2

To Identify Built-ins, Create Dispatch Table, Execute In-Process Commands, Handle Invalid Commands, Maintain State, Test Dispatch Logic

```
j_rakshitha@RAKSHITHA:~/ossp_project$ nano src/session9_task2_dispatch.c
j_rakshitha@RAKSHITHA:~/ossp_project$ gcc -Wall -Wextra -g src/session9_task2_dispatch.c -o build/session9_task2_dispatch
j_rakshitha@RAKSHITHA:~/ossp_project$ ./build/session9_task2_dispatch
=== Session 9 Task 2: Built-in Dispatch ===
Type 'help' for available commands.

dispatch-shell> help

Available built-ins:
  cd <directory>
  pwd
  echo <text>
  help
  exit

dispatch-shell> pwd
/home/j_rakshitha/ossp_project
dispatch-shell> echo Hello WSL
Hello WSL
dispatch-shell> cd /tmp
dispatch-shell> pwd
/tmp
dispatch-shell> abc
Invalid built-in command: abc
dispatch-shell> exit
Exit requested.
j_rakshitha@RAKSHITHA:~/ossp_project$ |
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task2_dispatch.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <unistd.h>

#define MAX_INPUT 1024
#define MAX_ARGS 64

typedef void (*builtin_function)(char **args);

typedef struct
{
    const char *name;
    builtin_function function;
} Builtin;

/* ----- Built-in functions ----- */

void builtin_pwd(char **args)
{
    char cwd[1024];

    (void)args;

    if (getcwd(cwd, sizeof(cwd)) != NULL)
        printf("%s\n", cwd);
    else
        perror("pwd");
}

void builtin_cd(char **args)
{
    if (args[1] == NULL)
    {
        const char *home = getenv("HOME");

        if (home == NULL)
        {
            fprintf(stderr, "cd: HOME not set\n");
            return;
        }
    }
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location
^X Exit	^R Read File	^_\ Replace	^U Paste	^J Justify	^/_ Go To Line

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task2_dispatch.c *
    }

    if (chdir(home) != 0)
        perror("cd");

    return;
}

if (chdir(args[1]) != 0)
    perror("cd");
}

void builtin_echo(char **args)
{
    for (int i = 1; args[i] != NULL; i++)
    {
        printf("%s", args[i]);

        if (args[i + 1] != NULL)
            printf(" ");
    }

    printf("\n");
}

void builtin_help(char **args)
{
    (void)args;

    printf("\nAvailable built-ins:\n");
    printf("  cd <directory>\n");
    printf("  pwd\n");
    printf("  echo <text>\n");
    printf("  help\n");
    printf("  exit\n\n");
}

void builtin_exit(char **args)
{
    (void)args;
|

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task2_dispatch.c *

    printf("Exit requested.\n");
    exit(0);
}

/* ----- Dispatch table ----- */

Builtin dispatch_table[] =
{
    {"cd", builtin_cd},
    {"pwd", builtin_pwd},
    {"echo", builtin_echo},
    {"help", builtin_help},
    {"exit", builtin_exit}
};

#define BUILTIN_COUNT \
    (sizeof(dispatch_table) / sizeof(dispatch_table[0]))

/* ----- Parse command ----- */

int parse_command(char *input, char **args)
{
    int count = 0;

    char *token = strtok(input, " \\t");

    while (token != NULL && count < MAX_ARGS - 1)
    {
        args[count++] = token;
        token = strtok(NULL, " \\t");
    }

    args[count] = NULL;

    return count;
}

/* ----- Dispatch command ----- */

int dispatch_command(char **args)

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\\ Replace   ^U Paste     ^J Justify   ^/ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~/o × + ▾
GNU nano 8.7.1 src/session9_task2_dispatch.c *
    if (args[0] == NULL)
        return 1;

    for (size_t i = 0; i < BUILTIN_COUNT; i++)
    {
        if (strcmp(args[0], dispatch_table[i].name) == 0)
        {
            dispatch_table[i].function(args);
            return 1;
        }
    }

    return 0;
}

int main(void)
{
    char input[MAX_INPUT];
    char *args[MAX_ARGS];

    printf("=== Session 9 Task 2: Built-in Dispatch ===\n");
    printf("Type 'help' for available commands.\n\n");

    while (1)
    {
        printf("dispatch-shell> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (parse_command(input, args) == 0)
            continue;

        if (!dispatch_command(args))
        {
            printf("Invalid built-in command: %s\n", args[0]);
        }
    }
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
    }

    return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Skill Session 10

Task 1

To Retrieve Current Directory, Display Path, Process Exit Requests, Cleanup Resources, Save State, Verify Correct Behavior

```
j_rakshitha@RAKSHITHA: ~  
j_rakshitha@RAKSHITHA:~$ nano session10_pwd_exit.c  
j_rakshitha@RAKSHITHA:~$ gcc -Wall -Wextra -g session10_pwd_exit.c -o session10_pwd_exit  
j_rakshitha@RAKSHITHA:~$ ./session10_pwd_exit  
myshell$ pwd  
/home/j_rakshitha  
myshell$ exit  
Exiting shell...  
j_rakshitha@RAKSHITHA:~$ |
```

```
GNU nano 8.7.1 session10_pwd_exit.c *  
#include <stdio.h>  
#include <stdlib.h>  
#include <unistd.h>  
#include <limits.h>  
#include <string.h>  
  
void show_pwd()  
{  
    char cwd[PATH_MAX];  
  
    if (getcwd(cwd, sizeof(cwd)) != NULL)  
    {  
        printf("%s\n", cwd);  
    }  
    else  
    {  
        perror("pwd");  
    }  
}  
  
int main()  
{  
    char command[100];  
  
    while (1)  
    {  
        printf("myshell$ ");  
        fflush(stdout);  
  
        if (fgets(command, sizeof(command), stdin) == NULL)  
        {  
            printf("\n");  
            break;  
        }  
  
        command[strcspn(command, "\n")] = '\0';  
  
        if (strcmp(command, "pwd") == 0)  
        {  
            show_pwd();  
        }  
        else if (strcmp(command, "exit") == 0)  
        {  
            printf("Exiting shell...\n");  
            break;  
        }  
        else if (strlen(command) == 0)  
        {  
            continue;  
        }  
        else  
        {  
            printf("Unknown command: %s\n", command);  
        }  
    }  
  
    return 0;  
}
```

^G Help	^O Write Out	^F Where Is	^K Cut	^T Execute	^C Location	M-U Undo
^X Exit	^R Read File	^I Replace	^U Paste	^J Justify	^_ Go To Line	M-E Redo

Task 2

To Parse Export Syntax, Update Environment Variables, Validate Names, Handle Existing Variables, Export to Child Processes, Test Functionality

```
j_rakshitha@RAKSHITHA: ~  
j_rakshitha@RAKSHITHA:~$ nano session10_export.c  
j_rakshitha@RAKSHITHA:~$ gcc -Wall -Wextra -g session10_export.c -o session10_export  
j_rakshitha@RAKSHITHA:~$ ./session10_export  
myshell$ export NAME=Rakshitha  
Exported: NAME=Rakshitha  
myshell$ env  
SHELL=/bin/bash  
WSL2_GUI_APPS_ENABLED=1  
WSL_DISTRO_NAME=Ubuntu  
NAME=Rakshitha  
PWD=/home/j_rakshitha  
LOGNAME=j_rakshitha  
HOME=/home/j_rakshitha  
LANG=C.UTF-8  
WSL_INTEROP=/run/WSL/334_interop  
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;32:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=01;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:*.zst=01;31:*.tzt=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;31:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;31:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=01;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.ppm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*.svg=01;35:*.svgz=01;35:*.mng=01;35:*.pcx=01;35:*.mov=01;35:*.mpg=01;35:*.mpeg=01;35:*.m2v=01;35:*.mkv=01;35:*.webm=01;35:*.webp=01;35:*.ogm=01;35:*.mp4=01;35:*.m4v=01;35:*.mp4v=01;35:*.vob=01;35:*.qt=01;35:*.nuv=01;35:*.wmv=01;35:*.asf=01;35:*.rm=01;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=01;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mka=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.opus=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.bak=00;90:*.old=00;90:*.orig=00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpg=00;90:*.dpg-dist=00;90:*.dpg-old=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00;90:*.rpmsave=00;90:  
WAYLAND_DISPLAY=wayland-0  
LESSCLOSE=/usr/bin/lesspipe %s %s  
TERM=xterm-256color  
LESSOPEN=| /usr/bin/lesspipe %s  
USER=j_rakshitha  
DISPLAY=:0  
SHLVL=1
```

```
j_rakshitha@RAKSHITHA: ~  
XDG_RUNTIME_DIR=/run/user/1000  
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com  
WSLENV=  
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop  
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII  
.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common F  
iles/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8  
path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/sy  
stem32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsP  
owerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA  
Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mn  
t/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Us  
ers/J Rakshitha/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/J Ra  
kshitha/AppData/Local/Programs/Python/Python314:/mnt/c/Users/J Rakshitha/AppData/Lo  
cal/Programs/Python/Launcher:/mnt/c/Users/J Rakshitha/.local/bin:/mnt/c/Users/J Ra  
kshitha/AppData/Local/Microsoft/WindowsApps:/mnt/d/Coding/gcc/bin:/mnt/c/Users/J Ra  
kshitha/AppData/Roaming/npm:/mnt/d/Coding/vs code/bin:/snap/bin  
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus  
HOSTTYPE=x86_64  
PULSE_SERVER=unix:/mnt/wslg/PulseServer  
_=/session10_export  
myshell$ export COLLEGE=KLU  
Exported: COLLEGE=KLU  
myshell$ env  
SHELL=/bin/bash  
WSL2_GUI_APPS_ENABLED=1  
WSL_DISTRO_NAME=Ubuntu  
NAME=Rakshitha  
PWD=/home/j_rakshitha  
LOGNAME=j_rakshitha  
HOME=/home/j_rakshitha  
LANG=C.UTF-8  
WSL_INTEROP=/run/WSL/334_interop  
LS_COLORS=rs=0:di=01;34:ln=01;36:mh=00:pi=40;33:so=01;35:do=01;35:bd=40;33;01:cd=40  
;33;01:or=40;31;01:mi=00:su=37;41:sg=30;43:ca=00:tw=30;42:ow=34;42:st=37;44:ex=01;3  
2:*.tar=01;31:*.tgz=01;31:*.arc=01;31:*.arj=01;31:*.taz=01;31:*.lha=01;31:*.lz4=01;  
31:*.lzh=01;31:*.lzma=01;31:*.tlz=01;31:*.txz=01;31:*.tzo=01;31:*.t7z=01;31:*.zip=0  
1;31:*.z=01;31:*.dz=01;31:*.gz=01;31:*.lrz=01;31:*.lz=01;31:*.lzo=01;31:*.xz=01;31:  
*.zst=01;31:*.tztst=01;31:*.bz2=01;31:*.bz=01;31:*.tbz=01;31:*.tbz2=01;31:*.tz=01;31  
:*.deb=01;31:*.rpm=01;31:*.jar=01;31:*.war=01;31:*.ear=01;31:*.sar=01;31:*.rar=01;3  
1:*.alz=01;31:*.ace=01;31:*.zoo=01;31:*.cpio=01;31:*.7z=01;31:*.rz=01;31:*.cab=01;3  
1:*.wim=01;31:*.swm=01;31:*.dwm=01;31:*.esd=01;31:*.avif=01;35:*.jpg=01;35:*.jpeg=0  
1;35:*.mjpg=01;35:*.mjpeg=01;35:*.gif=01;35:*.bmp=01;35:*.pbm=01;35:*.pgm=01;35:*.p  
pm=01;35:*.tga=01;35:*.xbm=01;35:*.xpm=01;35:*.tif=01;35:*.tiff=01;35:*.png=01;35:*
```

```
j_rakshitha@RAKSHITHA: ~  
1;35:*.rmvb=01;35:*.flc=01;35:*.avi=01;35:*.fli=01;35:*.flv=01;35:*.gl=01;35:*.dl=0  
1;35:*.xcf=01;35:*.xwd=01;35:*.yuv=01;35:*.cgm=01;35:*.emf=01;35:*.ogv=01;35:*.ogx=  
01;35:*.aac=00;36:*.au=00;36:*.flac=00;36:*.m4a=00;36:*.mid=00;36:*.midi=00;36:*.mk  
a=00;36:*.mp3=00;36:*.mpc=00;36:*.ogg=00;36:*.ra=00;36:*.wav=00;36:*.oga=00;36:*.op  
us=00;36:*.spx=00;36:*.xspf=00;36:*.~=00;90:*.bak=00;90:*.old=00;90:*.orig=  
00;90:*.part=00;90:*.rej=00;90:*.swp=00;90:*.tmp=00;90:*.dpgk-dist=00;90:*.dpgk-old  
=00;90:*.ucf-dist=00;90:*.ucf-new=00;90:*.ucf-old=00;90:*.rpmnew=00;90:*.rpmorig=00  
;90:*.rpmsave=00;90:  
WAYLAND_DISPLAY=wayland-0  
LESSCLOSE=/usr/bin/lesspipe %s %s  
TERM=xterm-256color  
LESSOPEN=| /usr/bin/lesspipe %s  
USER=j_rakshitha  
DISPLAY=:0  
SHLVL=1  
XDG_RUNTIME_DIR=/run/user/1000  
DEBUGINFOD_URLS=https://debuginfod.ubuntu.com  
WSLENV=  
XDG_DATA_DIRS=/usr/local/share:/usr/share:/var/lib/snapd/desktop  
PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/l  
ocal/games:/usr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII  
.WindowsSubsystemForLinux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Common F  
iles/Oracle/Java/javapath:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/java8  
path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java/javapath:/mnt/c/Windows/sy  
stem32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Windows/System32/WindowsP  
owerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files (x86)/NVIDIA  
Corporation/PhysX/Common:/mnt/c/Program Files/NVIDIA Corporation/NVIDIA NvDLISR:/mn  
t/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/MySQL Shell 8.0/bin:/mnt/c/Us  
ers/J Rakshitha/AppData/Local/Programs/Python/Python314/Scripts:/mnt/c/Users/J Ra  
kshitha/AppData/Local/Programs/Python/Python314:/mnt/c/Users/J Rakshitha/AppData/Lo  
cal/Programs/Python/Launcher:/mnt/c/Users/J Rakshitha/.local/bin:/mnt/c/Users/J Ra  
kshitha/AppData/Local/Microsoft/WindowsApps:/mnt/d/Coding/gcc/bin:/mnt/c/Users/J Ra  
kshitha/AppData/Roaming/npm:/mnt/d/Coding/vs code/bin:/snap/bin  
DBUS_SESSION_BUS_ADDRESS=unix:path=/run/user/1000/bus  
HOSTTYPE=x86_64  
PULSE_SERVER=unix:/mnt/wslg/PulseServer  
_=/session10_export  
COLLEGE=KLU  
myshell$ export 123NAME=test  
export: invalid variable name: 123NAME  
myshell$ export NAME  
export: invalid syntax  
Usage: export NAME=value  
myshell$ exit  
j_rakshitha@RAKSHITHA:~$ |
```

```
j_rakshitha@RAKSHITHA: ~  ×  +  ▾  -  □  ×
GNU nano 8.7.1 session10_export.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>

int valid_name(const char *name)
{
    if (name == NULL || name[0] == '\\0')
        return 0;

    if (!(isalpha((unsigned char)name[0]) || name[0] == '_'))
        return 0;

    for (int i = 1; name[i] != '\\0'; i++)
    {
        if (!(isalnum((unsigned char)name[i]) || name[i] == '_'))
            return 0;
    }

    return 1;
}

void handle_export(char *command)
{
    char *assignment = command + 7;

    while (*assignment == ' ')
        assignment++;

    if (*assignment == '\\0')
    {
        printf("export: missing argument\\n");
        return;
    }

    char *equal = strchr(assignment, '=');

    if (equal == NULL)
    {
        printf("export: invalid syntax\\n");
        printf("Usage: export NAME=value\\n");
    }
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~ × + ▾
GNU nano 8.7.1 session10_export.c *
    printf("Usage: export NAME=value\n");
    return;
}

*equal = '\0';

char *name = assignment;
char *value = equal + 1;

if (!valid_name(name))
{
    printf("export: invalid variable name: %s\n", name);
    return;
}

if (setenv(name, value, 1) == -1)
{
    perror("export");
    return;
}

printf("Exported: %s=%s\n", name, value);
}

int main()
{
    char command[1024];

    while (1)
    {
        printf("myshell$ ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "exit") == 0)
        {
            break;
        }
    }
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
j_rakshitha@RAKSHITHA: ~ × + v
GNU nano 8.7.1 session10_export.c *

while (1)
{
    printf("myshell$ ");
    fflush(stdout);

    if (fgets(command, sizeof(command), stdin) == NULL)
        break;

    command[strcspn(command, "\n")] = '\0';

    if (strcmp(command, "exit") == 0)
    {
        break;
    }

    if (strcmp(command, "env") == 0)
    {
        extern char **environ;

        for (char **env = environ; *env != NULL; env++)
            printf("%s\n", *env);

        continue;
    }

    if (strncmp(command, "export ", 7) == 0)
    {
        handle_export(command);
        continue;
    }

    if (strlen(command) == 0)
        continue;

    printf("Unknown command: %s\n", command);
}

return 0;
}

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```